

Delphes event selection vs. CMS HIG-25-008

what the ntuple converter applies, and what it cannot

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Abstract

Comparison of the CMS Run-3 $HH \rightarrow b\bar{b}\tau\tau$ selection (HIG-25-008, Table 1 and Sections 5–6) against the selection applied to the Delphes samples. The purpose is to make every difference explicit, so that yields and sensitivities computed on the Delphes samples are read with the right caveats rather than compared naively against the published numbers.

1 Scope

The “Delphes” column throughout describes the HIG-25-008 resolved selection as implemented in `scripts/delphes_to_sbi.py`, invoked with `--cms-selection`. **It is not the default.** Without that flag the converter applies a deliberately loose preselection — one light lepton, one τ_h , two jets, with no b-tag requirement, no charge requirement, no vetoes and no mass windows — which exists to mirror the inputs of the NSBI test rather than the CMS analysis, and whose signal/ $t\bar{t}$ acceptance ratio is correspondingly ~ 2.5 rather than the hundreds a real $b\bar{b}\tau\tau$ selection gives. Numbers quoted from Delphes should always state which of the two was used.

Both modes are semi-leptonic by default ($\tau_\mu\tau_h$, $\tau_e\tau_h$), matching the scope of the NSBI test; `--channels mt,et,tt` adds $\tau_h\tau_h$, CMS’s most sensitive channel.

2 Event content and channels

	CMS HIG-25-008	Delphes
Channels	$\tau_\mu\tau_h$, $\tau_e\tau_h$, $\tau_h\tau_h$, boosted- $\tau_h\tau_h$	$\tau_\mu\tau_h$, $\tau_e\tau_h$ ✓; $\tau_h\tau_h$ opt-in
Channel assignment	$\mu \rightarrow \tau_\mu\tau_h$, else $e \rightarrow \tau_e\tau_h$, else 2nd $\tau_h \rightarrow \tau_h\tau_h$ (§5)	same priority ✓
Trigger	single- ℓ , cross ℓ - τ , di- τ , di- τ +jet, HH-Parking, VBF, boosted (§5)	✗ none
Categories	boosted / VBF / resolved 1b1j / re- solved 2b $\rightarrow$ 10 SRs (§6)	✗ none
Discriminant	ggHH and qqHH DNNs, 5-fold en- semble (§6)	— (NSBI replaces it)

3 Object selection

4 Pair selection and signal region

The elliptical selections implemented, from Eqs. 2–4:

$$\begin{aligned} \frac{(m_{\tau\tau} - 116)^2}{61^2} + \frac{(m_{bb} - 114)^2}{228^2} &< 1 & (\tau_\mu\tau_h) \\ \frac{(m_{\tau\tau} - 119)^2}{57^2} + \frac{(m_{bb} - 109)^2}{232^2} &< 1 & (\tau_e\tau_h) \\ \frac{(m_{\tau\tau} - 117)^2}{54^2} + \frac{(m_{bb} - 134)^2}{169^2} &< 1 & (\tau_h\tau_h) \end{aligned}$$

CMS quotes $\sim 99\%$ signal efficiency for these, with background efficiencies of 62%, 60% and 54% respectively.

	CMS (Table 1)	Delphes
Muon p_T	$> 22/26/6$ GeV (cross μ - τ / single μ / VBF+ μ)	> 22 (cross) ✓
Muon $ \eta $	< 2.4	≤ 2.4 ✓
Muon ID, isolation	Tight Muon ID, PF rel. iso < 0.15	✗ none
Electron p_T	$> 25/31/13(18)$ GeV (cross e - τ / single e / VBF+ e)	> 25 (cross) ✓
Electron $ \eta $	< 2.5	≤ 2.5 ✓
Electron ID	Tight MVA Isolation+ID (80% eff.)	✗ none
Lepton $ d_{xy} , d_z $	$< 0.045, < 0.2$ cm	✗ not available
τ_h p_T	32 ($\tau_\mu\tau_h$), 35 ($\tau_e\tau_h$), 40/35/25 ($\tau_h\tau_h$, trigger-dependent)	32 / 35 ✓; 40 ($\tau_h\tau_h$) ✓
τ_h $ \eta , d_z $	$< 2.5, < 0.2$ cm	≤ 2.5 ✓, no d_z
τ_h identification	DEEPTAU: D_{jet}^τ Medium, D_e^τ VVLoose, D_μ^τ VLoose/Tight	✗ one tag bit
τ_h object	HPS; decay modes 1-prong+0, 1, 2 π^0 , 3-prong+0, 1 π^0	✗ AK4 jet with a tag bit
$H \rightarrow b\bar{b}$ jets	AK4, $p_T > 20$, $ \eta < 2.5$, $\Delta R(j, \ell) > 0.5$ from both τ candidates (§5)	✓ exactly as CMS
b-jet pair assignment	HH-BTAG DNN (§5)	✗ two highest b-tag
b-tagging	UPART (2024); categories split at the 10% mistag WP (§6)	--btag-min at UPART Medium ($\sim 1\%$ mistag)

	CMS	Delphes
Opposite charge	required (Table 1)	✗ no — see §5
$\Delta R(\ell, \tau_h)$	> 0.5 (resolved); < 0.8 (boosted)	> 0.5 ✓
Additional-lepton veto	required (Table 1, §5)	yes ✓
Signal region	ellipse in $(m_{\tau\tau}, m_{bb})$, Eqs. 2–4	yes ✓
$m_{\tau\tau}$ used in the ellipse	DNN-regressed $\tau\tau$ mass (§5)	FastMTT (closest available)

5 What Delphes cannot support, and why

Opposite charge. The ntuple’s `Jet` collection carried no charge field, so the τ_h leg had no charge to compare and `--cms-selection` is charge-blind. This matters more than it sounds: same-sign pairs are CMS’s QCD control region at $\sim 99\%$ purity (§7), i.e. an essentially pure fake- τ population that CMS removes and we keep. *Delphes does write `Jet.Charge`* — it was simply not carried into the ntuple schema. Now added; it requires a re-ntuplisation (not a re-simulation) to take effect.

Lepton ID, isolation, impact parameters. Delphes leptons carry only the card’s efficiency parameterisation. There is no ID discriminant, no isolation and no d_{xy} or d_z to cut on.

τ_h object and identification. A Delphes τ_h *is* an AK4 jet carrying a single tag bit, not an HPS-reconstructed visible τ . Consequences: jet-level four-vector (underlying event inside $R = 0.4$), no decay modes, and one bit in place of the three DEEPTAU discriminants. This shifts the whole $\tau\tau$ block — $m_{\tau\tau}$, $\Delta R_{\tau\tau}$, and the ellipse acceptance — and no selection change addresses it.

b-jet pair assignment. CMS uses the HH-BTAG DNN; we take the two highest-b-tag jets. This lands directly on m_{bb} and ΔR_{bb} .

b-tag working point. The Delphes bit is UPART-AK4 Medium (cut 0.1272; card PATCH-6), i.e. $\sim 1\%$ mistag. CMS’s resolved 1b1j/2b split uses the **10% mistag** working point. `--btag-min 2` is therefore *tighter* than CMS’s 2b category boundary, not equivalent to it.

Trigger and categorisation. No trigger is applied, and the sample is flat rather than split into the ten category $\times$ channel signal regions CMS fits.

Boosted and VBF topologies. AK8 jets and the VBF jet pair are not written to the ntuple, so those categories are absent entirely.

6 Consequences worth stating alongside any number

1. **The selection described here is not the converter’s default.** It requires `--cms-selection`; the default preselection is much looser and must never be compared directly against published yields.
2. **m_{HH} and $m_{\tau\tau}$ are visible masses** in the converter output. The CMS reference ntuple takes m_{HH} from a branch (`mass_tautau`) whose definition should be confirmed — if it is the corrected $\tau\tau$ system, the two are different observables and any m_{HH} threshold behaves differently on each. FastMTT-corrected versions (`m_hh_fastmtt`, `m_tautau_fastmtt`) are carried alongside for exactly this reason.
3. **The stock (untuned) card underestimates the $j \rightarrow \tau_h$ fake rate.** CMS quotes 10^{-2} – 10^{-3} (§7); in the untuned samples the fully-hadronic $t\bar{t}$ component contributes essentially nothing. The fake background is therefore *too small*, and S/B correspondingly optimistic. This is a detector-model deficiency, not a selection one, and is what the tuning τ -mistag map addresses.
4. **$\Delta R(j, \ell)$ differs between the two modes (0.4 vs 0.5).** Small, but it lands on ΔR_{bb} and m_{bb} , both of which are training features.

Implementation: `scripts/delphes_to_sbi.py` — `cms_select()` (CMS selection), `select()` (preselection), `_in_ellipse()` (Eqs. 2–4), with the Table 1 thresholds and ellipse coefficients pinned by tests in `tests/test_delphes_to_sbi.py`.